

Replication Package Data Directory Guide

“Structural Transformation Beyond Productivity: Subsistence Organization and Agricultural Commercialization in the United States”

Separate data-directory documentation for the replication repository

Replication repository:

<https://github.com/hanwang114-droid/crop-subsistence-snap-replication>

This document explains how to reconstruct the raw-data directory required to reproduce the analyses, calibration procedures, figures, and tables in the replication package. It is intended as a standalone data-directory guide, separate from the main repository README.

The replication workflow stores original downloaded source files in raw_data/. The script 01_clean_raw_data.py reads those original files and writes cleaned analysis files used by the later scripts. To preserve provenance, the filenames listed below retain the original names from the downloaded source files whenever possible

Recommended Raw-Data Directory Structure

```
raw_data/
├── income/
├── ipums/
│   ├── code_book/
│   ├── command_files/
│   └── data/
├── population/
├── quickstats/
├── snap/
├── usda/
├── census_gazetteer/
└── nhgis/
```

Required Raw Data Files

Folder	Files to save using original downloaded names	Purpose in replication workflow
raw_data/ipums/	usa_00001.dat usa_00001.xml usa_00001.txt R or Stata loading/command files IPUMS USA_extract summary(1).pdf	IPUMS USA extract used to construct agricultural, forestry, and fishing labor-force shares by state and census year.
raw_data/usda/	table01 Table02StateProductivityAccounts	USDA ERS national TFP series and state agricultural productivity, output, and labor-input accounts.
raw_data/income/	household-income-2026-02	Farm household income ratio used in the motivation regression.
raw_data/quickstats/	79C83B62-EF7A-3A1D-8915-FC78A71F9EEF (corn) 0602491F-BA1F-3E58-9C29-E27D50A7AA6B (cotton) 40ACD0F9-6354-35D2-9B22-597BA369FA7E (soybeans) C51AF319-6449-3723-928F-3F64137B063D (suitability)	Historical crop acreage and fieldwork suitability measures from USDA NASS Quick Stats.
raw_data/snap/	snap-persons-5	State SNAP persons used to construct SNAP participation rates.
raw_data/population/	NST-EST2025-POP	July 1 resident population denominator used to construct SNAP participation rates.
raw_data/nhgis/	nhgis0002_ds210_1920_county.csv; nhgis0002_ds210_1920_county_codebook.txt	IPUMS NHGIS 1920 Census of Agriculture county extract; AB4X001 soybean acreage in 1919 used to construct log(1 + soybean acres per 1,000 land acres).
N/A	Census API key	Census API key used to access ACS 2024 5-year county-level variables through the Census Data API. Not a data source itself; it is an access credential for downloading Census API data.
raw_data/census_gazetteer/	census_2024_county_gazetteer.zip	Census 2024 county Gazetteer file; land area used to construct land acres, log land acres, and soybean acres per 1,000 land acres.

Data Sources and Citation: Information

1. IPUMS USA

Source: IPUMS USA, University of Minnesota. The replication package uses a custom IPUMS USA extract to construct agricultural, forestry, and fishing employment shares. The extract summary should be retained with the raw files so that the sample composition, selected variables, and case-selection criteria remain transparent.

- Required files: usa_00001.dat, usa_00001.xml, usa_00001.txt, R or Stata loading/command files, and IPUMS USA_extract summary(1).pdf.
- Key variables: YEAR, STATEFIP, PERWT, LABFORCE, and IND1950.
- The analysis uses LABFORCE to define the labor-force denominator, PERWT for person weighting, and IND1950 codes 105, 116, and 126 for agriculture, forestry, and fishing employment.

Citation: Steven Ruggles, Sarah Flood, Matthew Sobek, Daniel Backman, Grace Cooper, Julia A. Rivera Drew, Stephanie Richards, Renae Rodgers, Jonathan Schroeder, and Kari C.W. Williams. *IPUMS USA: Version 16.0* [dataset]. Minneapolis, MN: IPUMS, 2025. <https://doi.org/10.18128/D010.V16.0>

2. USDA ERS Agricultural Productivity in the United States

Source: United States Department of Agriculture, Economic Research Service (ERS), Agricultural Productivity in the United States.

- Required files: table01 and Table02StateProductivityAccounts.
- table01 provides the national agricultural total factor productivity series used in the motivation analysis.
- Table02StateProductivityAccounts provides state-level productivity, output, input, and labor-input measures used for structural calibration.

Citation: United States Department of Agriculture, Economic Research Service. Agricultural Productivity in the United States. <https://www.ers.usda.gov/data-products/agricultural-productivity-in-the-united-states>

3. USDA ERS Farm Household Income and Characteristics

Source: United States Department of Agriculture, Economic Research Service (ERS), Farm Household Income and Characteristics.

- Required file: household-income-2026-02.
- Required series: Farm Household Financial Indicators, Mean and median U.S. farm operator household income and ratio of farm household to U.S. household income, 1960-2024.

Citation: United States Department of Agriculture, Economic Research Service. Farm Household Income and Characteristics. <https://www.ers.usda.gov/data-products/farm-household-income-and-characteristics>

4. USDA NASS Quick Stats

Source: United States Department of Agriculture, National Agricultural Statistics Service (NASS), Quick Stats Database.

- Required crop exports: 79C83B62-EF7A-3A1D-8915-FC78A71F9EEF (corn), 0602491F-BA1F-3E58-9C29-E27D50A7AA6B (cotton), and 40ACD0F9-6354-35D2-9B22-597BA369FA7E (soybeans).
- Crop query settings: Program = SURVEY; Sector = CROPS; Group = FIELD CROPS; Category = AREA PLANTED; Data Item = CORN/COTTON/SOYBEANS - ACRES PLANTED; Geographic Level = STATE.
- Required suitability export: C51AF319-6449-3723-928F-3F64137B063D (suitability).

- Suitability query settings: Program = SURVEY; Sector = CROPS; Group = CROP TOTALS; Commodity = FIELDWORK; Category = DAYS SUITABLE; Data Item = MEASURED IN DAYS / WEEK; Geographic Level = STATE.

Citation: United States Department of Agriculture, National Agricultural Statistics Service. Quick Stats Database.
<https://quickstats.nass.usda.gov/>

5. USDA FNS SNAP Participation Data

Source: United States Department of Agriculture, Food and Nutrition Service (FNS), Supplemental Nutrition Assistance Program Participation and Costs.

- Required file: snap-persons-5.
- Required table: State Level Participation and Benefits, latest available month, Persons.
- Purpose: provides the numerator for state SNAP participation rates.

Citation: United States Department of Agriculture, Food and Nutrition Service. Supplemental Nutrition Assistance Program Participation and Costs. <https://www.fns.usda.gov/pd/supplemental-nutrition-assistance-program-snap>

6. U.S. Census Bureau Population Estimates

Source: United States Census Bureau, Population Estimates Program.

- Required file: NST-EST2025-POP.
- Required table: Annual Estimates of the Resident Population for the United States, Regions, States, District of Columbia and Puerto Rico: April 1, 2020 to July 1, 2025.
- Purpose: provides the denominator for state SNAP participation rates.

Citation: United States Census Bureau. Population Estimates Program. Annual Estimates of the Resident Population for the United States, Regions, States, District of Columbia and Puerto Rico: April 1, 2020 to July 1, 2025.
<https://www.census.gov/data/tables/time-series/demo/popest/2020s-state-total.html>

7. IPUMS NHGIS 1920 Census of Agriculture County Extract

Source: IPUMS NHGIS, University of Minnesota. The replication package uses a custom NHGIS county-level extract from the 1920 Census of Agriculture to construct historical soybean intensity at the county level.

- Required directory: raw_data/nhgis/
- Required files: nhgis0002_ds210_1920_county.csv and nhgis0002_ds210_1920_county_codebook.txt.
- Required table: 1920 Census of Agriculture county table containing crop acreage by type.
- Key variable: AB4X001, soybean acreage in 1919.
- Purpose: used to construct $\log(1 + \text{soybean acres per 1,000 county land acres})$ for the county-level reduced-form regressions.

Citation: Steven Manson, Jonathan Schroeder, David Van Riper, Tracy Kugler, and Steven Ruggles. IPUMS National Historical Geographic Information System: Version 19.0 [dataset]. Minneapolis, MN: IPUMS, 2024.
<https://doi.org/10.18128/D050.V19.0>

8. U.S. Census Bureau ACS 2024 Five-Year County Data

Source: United States Census Bureau, American Community Survey 2024 five-year estimates, accessed through the Census Data API.

- Access credential: Census API key.
- Note: the Census API key is not a data source itself. It is an access credential used to download public ACS

county-level variables through the Census Data API.

- Purpose: provides county-level SNAP household receipt rates and county socioeconomic, demographic, and population controls used in the county-level reduced-form regressions.
- Main outcome: county SNAP household receipt rate.
- Controls: population, racial and ethnic composition, educational attainment, median household income, poverty rate, and unemployment rate.

Citation: United States Census Bureau. American Community Survey 2024 Five-Year Data. Census Data API.

<https://www.census.gov/data/developers/data-sets/acs-5year.html>

9. U.S. Census Bureau County Gazetteer File

Source: United States Census Bureau, Gazetteer Files.

- Required directory: `raw_data/census_gazetteer/`
- Required file: `census_2024_county_gazetteer.zip`.
- Required table: 2024 county Gazetteer file.
- Purpose: provides county land area used to construct land acres, log land acres, and soybean acres per 1,000 county land acres.

Citation: United States Census Bureau. Gazetteer Files. 2024 County Gazetteer File.

<https://www.census.gov/geographies/reference-files/time-series/geo/gazetteer-files.html>

Notes on Data Distribution

Raw data files are not distributed directly through the GitHub repository. This document records the original data providers, original downloaded filenames, query settings, and directory locations needed to reconstruct the raw-data directory.